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1. a. $\begin{cases} \ln(C_0)+\ln(C_1)+\ln(C_2) \\ Y-(P_0C_0+P_1C_1+P_2C_2)\geq 0 \end{cases}$

$f(C_0,C_1,C_2,\lambda)=\ln(C_0)+\ln(C_1)+\ln(C_2)+\lambda(Y-(P_0C_0+P_1C_1+P_2C_2))$

$\frac{\partial f}{\partial C_0}=\frac{1}{C_0}-\lambda P_0$, $\frac{\partial f}{\partial C_1}=\frac{1}{C_1}-\lambda P_1$, $\frac{\partial f}{\partial C_2}=\frac{1}{C_2}-\lambda P_2$, $\frac{\partial f}{\partial \lambda}=Y-(P_0C_0+P_1C_1+P_2C_2)$

b. The constraint bind

because $\begin{cases} \frac{\partial f}{\partial C_0}=0 \\ \frac{\partial f}{\partial C_1}=0 \\ \frac{\partial f}{\partial C_2}=0 \\ \frac{\partial f}{\partial \lambda}=0 \end{cases}$

c. $C_0=\frac{1}{\lambda P_0}$ $C_1=\frac{1}{\lambda P_1}$ $C_2=\frac{1}{\lambda P_2}$

$Y-\frac{3}{\lambda}=0$ $\therefore \lambda=\frac{3}{Y}$ $\therefore$ when $C_0=\frac{Y}{3P_0}$ $C_1=\frac{Y}{3P_1}$ $C_2=\frac{Y}{3P_2}$,

$\max_{C_0,C_1,C_2} \ln(C_0)+\ln(C_1)+\ln(C_2)=\ln\frac{Y^3}{27P_1P_2P_3}$

d.

good	0	1	2
	C_0	C_1	C_2
unit price	P_0	P_1	P_2

$\therefore$ good 0: $P_0C_0=\frac{Y}{3}$ $\therefore$ 33.3%

1: $P_1C_1=\frac{Y}{3}$ 33.3%

2: $P_2C_2=\frac{Y}{3}$ 33.3%

2. a. $g(x)=y-K^aL^b\leq 0$

$\mathcal{L}(y,K,L,\lambda)=py-rk-wL+\lambda(y-K^aL^b)$

b. $\frac{\partial \mathcal{L}}{\partial y}=p+\lambda=0$ $\lambda^*=-p$

$\frac{\partial \mathcal{L}}{\partial k}=-r-a\lambda L^bK^{a-1}=0$ $(K^*)^{a-1}(L^*)^b a\lambda^*=-r$ $\therefore (K^*)^{a-1}(L^*)^b=\frac{r}{ap}$

$\frac{\partial \mathcal{L}}{\partial L}=-w-b\lambda K^aL^{b-1}=0$ $\therefore (L^*)^{b-1}(K^*)^a\lambda^* b=w$ $\therefore (K^*)^a(L^*)^{b-1}=\frac{w}{pb}$

$\frac{\partial \mathcal{L}}{\partial \lambda}=y-K^aL^b=0$ $\therefore (K^*)^a(L^*)^b=y^*$ $\therefore \frac{L^*}{K^*}=\frac{rb}{aw}$

c. $\therefore L^*=\frac{br}{aw}K^*$ $\left(\frac{br}{aw}\right)^b K^{a+b-1}=\frac{r}{ap}$

$\therefore K^*=\left[\left(\frac{r}{ap}\right)\left(\frac{br}{aw}\right)^{-b}\right]^{\frac{1}{a+b-1}}$

$L^*=\left[\left(\frac{r}{ap}\right)\left(\frac{br}{aw}\right)^{-b}\right]^{\frac{1}{a+b-1}}\cdot\frac{br}{aw}$ $\lambda^*=-p$

$y^*=\left[\left(\frac{r}{ap}\right)\left(\frac{br}{aw}\right)^{-b}\right]^{\frac{a+b}{a+b-1}}\left(\frac{br}{aw}\right)^b$

d. i) p rises, y^* rises, k^* rises, l^* rises $\quad a+b-1<0$

ii) $\frac{(a+b)(1-b)}{a+b-1}+b=\frac{a}{a+b-1}<0 \quad \frac{1-b}{a+b-1}<0 \quad \frac{1-b}{a+b-1}+1=\frac{a}{a+b-1}<0$

$\therefore$ r rises, y^* falls. k^* falls. l^* falls

iii) $\frac{b}{a+b-1}<0 \quad \frac{(a+b)b}{a+b-1}-b=\frac{b}{a+b-1}<0 \quad \frac{b}{a+b-1}-1=\frac{b-a-b+1}{a+b-1}=\frac{1-a}{a+b-1}<0$

$\therefore$ w rises, y^* falls, k^* falls, l^* falls.

iv) $p\to 2p$, $r\to 2r$, $w\to 2w$

$y_1^*=\left[\frac{r}{ap}\left(\frac{br}{aw}\right)^{-b}\right]^{\frac{a+b}{a+b-1}}\cdot\left(\frac{br}{aw}\right)^{b}=y^*$, same

$k_1^*=\left[\left(\frac{r}{ap}\right)\left(\frac{br}{aw}\right)^{-b}\right]^{\frac{1}{a+b-1}}=k^*$, same

$l_1^*=\left[\left(\frac{r}{ap}\right)\left(\frac{br}{aw}\right)^{-b}\right]^{\frac{1}{a+b-1}}\cdot\frac{br}{aw}=l^*$, same

3. a. $E[u(x)]=\int u(x)f(x)dx=\int -e^{-vx}\cdot\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}dx=\frac{-1}{\sigma\sqrt{2\pi}}\int e^{-\left[\frac{(x-\mu)^2}{2\sigma^2}+vx\right]}dx$

$u(E[x])=-e^{-v\mu} \qquad u'(x)=ve^{-vx},\ v>0 \qquad \therefore u'(x)>0,\ u''(x)=-v^2e^{-vx}<0$

$\therefore u(x)$ is concave

$E[u(x)]<u(E(x))$

b. $E[u(ax)]=\int u(ax)f(x)d(ax)=\int e^{-vax}\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}dax$

$=\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{1}{2}\frac{(x-\mu+\sigma^2av)^2}{\sigma^2}-\mu av+\frac{\sigma^2a^2v^2}{2}\right)dx$

$=-\exp\left(-\mu av+\frac{\sigma^2a^2v^2}{2}\right)\cdot\frac{1}{\sqrt{2\pi\sigma^2}}\int\exp\left(-\frac{1}{2}\frac{(x-\mu+\sigma^2av)^2}{\sigma^2}\right)dx$

$=-\exp\left(-\mu av+\frac{\sigma^2a^2v^2}{2}\right)$

c. $\because E[u(ax)]=-\exp\left(-\mu av+\frac{\sigma^2a^2v^2}{2}\right)$

$E[u(ax)]_{max}\iff\left[-\mu av+\frac{\sigma^2a^2v^2}{2}\right]_{min}\iff\left[\left(\mu a-\frac{\sigma^2a^2v}{2}\right)v\right]_{max}$

$\because v>0$ and is a parameter $\quad\therefore \max_a E[u(ax)]$ is the same as $\max_a\left[\mu a-\frac{1}{2}v\sigma^2a^2\right]$